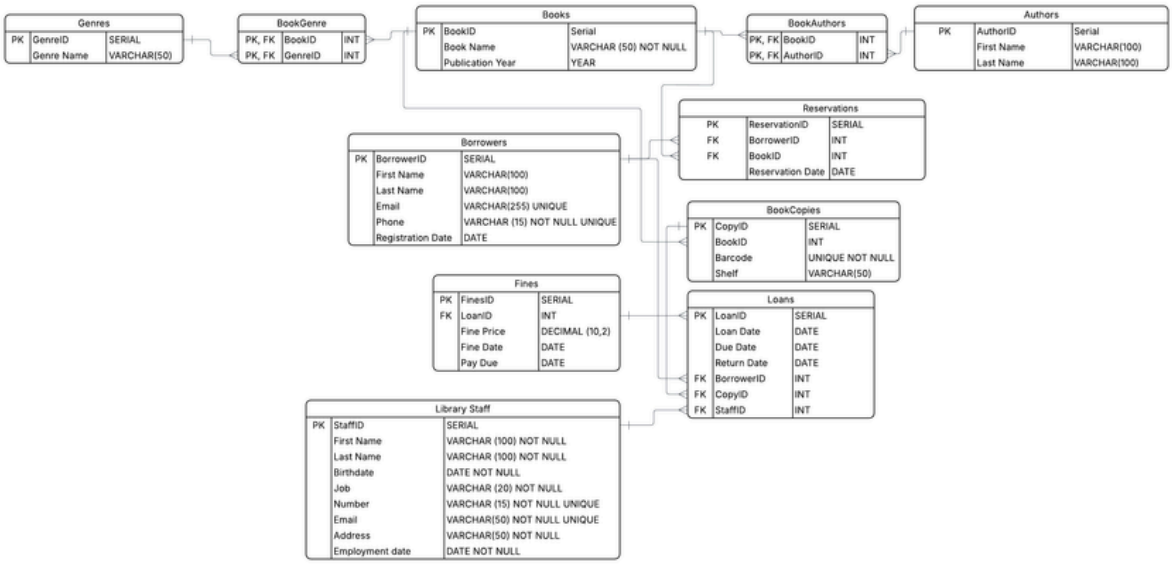
1.1 Logical Scheme



1.2 Objects

Books

This table stores the core information for every book title in the library catalog.

Table Name	Field Name	Field Description	Data Type / Constraints
Books	BookID	Unique indentifier for the book title	SERIAL
	Book Name	Title of the book	VARCHAR(50), NOT NULL
	Publication Year	The year the book was released	YEAR

One to many with **BookCopies**: one book title can have multiple physical copies

Many to many with **Authors** and **Genres** using bridge tables

Example with data

BookID	Book Name	Publication Year
1	The Great Gatsby	1925

Authors

This table stores information about the creators of the books in the library system.

Table Name	Field Name	Field Description	Data Type / Constraints
Authors	AuthorID	Reference to the book	SERIAL
	First Name	Author's first name	VARCHAR(50), NOT NULL
	Last Name	Author's last name	VARCHAR(50), NOT NULL

Many to many with **Books** through the **BookAuthors** bridge table: one author can write many books, and one book can have multiple authors

Example with data

AuthorID	First Name	Last Name
105	Scott	Fitzgerald

BookAuthors

This table resolves the many to many relationship between book and authors.

Table Name	Field Name	Field Description	Data Type / Constraints
BookAuthors	BookID	Reference to the book	INT, PK, FK
	AuthorID	Reference to the author	INT, PK, FK

Uses a composite primary key to ensure a specific author isn't linked to the same book twice

Example with data

BookID	AuthorID
1	105

Genres

A lookup table, like a dictionary, of all literary categories available in the library.

Table Name	Field Name	Field Description	Data Type / Constraints
Genres	GenreID	Unique identifier for the genre	SERIAL
	Genre Name	The name of the category	VARCHAR(50), NOT NULL, UNIQUE

Many to many with **Books** through the **BookGenre** bridge table

Example with data

BookID	AuthorID
5	Science Fiction
6	Classic Literature

BookGenre

This junction table resolves the many to many relationship between books and their literary genres

Table Name	Field Name	Field Description	Data Type / Constraints
BookGenre	BookID	Reference to the book title	INT, PK, FK
	GenreID	Reference to the genre category	INT, PK, FK

Also uses a composite primary key to ensure a book is never assigned to the same genre more than once

Example with data

BookID	GenreID
1	6

Library Staff

This table stores the details of the employees who manage the library operations, loans and returns

Table Name	Field Name	Field Description	Data Type / Constraints
Library Staff	StaffID	Unique identifier for the staff member	SERIAL
	First Name	Staff member's first name	VARCHAR(100), NOT NULL
	Last Name	Staff member's last name	VARCHAR(100), NOT NULL
	Birthdate	Employee's date of birth	DATE, NOT NULL
	Job	Specific job title or role	VARCHAR(20), NOT NULL
	Number	Contact phone number	VARCHAR(15), NOT NULL, UNIQUE
	Email	Professional email address	VARCHAR(50), NOT NULL, UNIQUE
	Address	Residential address	VARCHAR(50), NOT NULL
	Employment date	Date the staff member started	DATE, NOT NULL

One to many with **Loans** - one staff member can be responsible for authorizing and processing many different loan transactions.

Example with data

StaffID	First Name	Last Name	Birthdate	Job	Number	Email	Address	Employment date
50	Jane	Smith	1985-05-15	Librarian	+77777777777	j.smith@gmail.com	123 Maple St	2026-01-10

Borrwers

This table stores the details of the borrwer who loan and return books from library.

Table name	Field Name	Field Description	Data Type / Constraints
Borrwers	Borrwer_id	Unique identifier for the borrwer	SERIAL
	First_Name	Borrwer's first name	VARCHAR(100)
	Last_Name	Borrwer's last name	VARCHAR(100)
	Email	Active email address	VARCHAR(255)
	Phone	Contact phone number	VARCHAR(15) NOT NULL UNIQUE
	Registration_Date	Date of register	DATE

An idividual table that is connected to other tables as forien key like Loans, FInes and etc.

Example with data

Borrwer_ID	First Name	Last Name	Email	Number	Registration_date
12	Bob	Lane	bob_lane@gmail.com	+77777777777	2026-02-18

Reservations

A table for all reserved places in the librery.

Table name	Field Name	Field Description	Data Type / Constraints
Reservations	Reservation_id	Unique identifier for the reservation	SERIAL
	Borrwer_id	Reference to the borwer	INT, FK
	Book_Id	Reference to the book	INT, FK
	Reservation_Date	Date of reservation	DATE

This table connected to **Borrwers** as one to one and one to many for **Books**.

Example with data

Reservation_ID	Borrwer_id	Book_Id	Reservation_date
5	45	78	2026-04-13

Book copies

Every copy of the book and the shelf they located in.

Table name	Field Name	Field Description	Data Type / Constraints
Book_copies	Copy_Id	Unique identifier for the book's copy	SERIAL
	Book_id	Reference to the book	INT, FK
	Barcode	Unique code on book of the book	INT
	Shelf	The name of shelf the book is located	VARCHAR(50)

Book_copies only to **Books** as many to one.

Example with data

Copy_ID	Book_id	Barcode	Shelf
8	154	9 785171 583309	Horror

Fines

This table contains every fine that borrwer got from damiging or returning the over due book.

Table name	Field Name	Field Description	Data Type / Constraints
Fines	Fines_Id	Unique identifier for the fine	SERIAL
	Loans_id	Reference to the loans	INT, FK
	Fine Price	Price of fine	DECIMAL (10, 2)
	Fine Date	The date fine was registered	DATE
	Pay Due	The last date to pay the fine	DATE

Fines table only connected to **Loans** as one to one.

Example with data

Fine_ID	Loan_id	Fine Price	Fine Date	Pay Due
85	114	5,000	2026-01-13	2026-03-13

Loans

The table that stores every loaned book borrwers take with them.

Table name	Field Name	Field Description	Data Type / Constraints
Loans	Loan_Id	Unique identifier for the loan	SERIAL
	Loan Date	Reference to the loans	INT, FK
	Due Date	Last day for book to returned brfore any fine	DATE
	Return Date	Date of returned book	DATE
	Borrwer_Id	Reference to the borrwer	INT, FK
	Copy_Id	Reference to the books copy	INT, FK
	Staff_Id	Reference to the staff	INT, FK

Loans connected to more than one table. Such as **Borrwers** as many to one, **Staff** as many to one and **Book_Copy** many to one.

Example with data

Loan_ID	Loan Date	Due Date	Return Date	Borrwer_Id	Copy_Id	Staff_Id
58	2026-01-05	2026-02-05	2026-01-13	110	12	33